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Georgios Bardanikas

Machine Learning ▪ Signal Processing ▪ Neural Data Analysis

Computational neuroscientist with 4+ years of experience developing machine learning and signal processing approaches for large-scale neural and behavioral datasets. Experienced in designing reproducible end-to-end analysis pipelines and evaluating model robustness across high-dimensional heterogeneous biological datasets.

Technical Skills

Programming	Python, PyTorch, LaTeX, Git, Matlab
Machine Learning	RNNs, temporal CNNs, reinforcement learning, classification, clustering
Data Analysis	State-space dynamics, bayesian modeling, dynamical systems, information theory
Signal Processing	Time-series analysis, spectral methods, functional connectivity
Data Engineering	Data pipelines, preprocessing, reproducible workflows

Experience

02/2026 – Present
Computational Neuroscientist
Institut de Neurosciences de la Timone, Marseille, France

- Applied machine learning algorithms to characterize large-scale brain network dynamics from intracranial recordings across 148 brain regions in 24 epilepsy patients.

10/2021 – 12/2025
PhD Researcher – Computational Systems Neuroscience
Institut de Neurosciences de la Timone, Marseille, France

- Designed and executed 3 end-to-end analysis projects on large-scale neural and behavioral datasets.
- Developed and evaluated machine learning models to decode behavioral states from high-dimensional neural recordings across heterogeneous experimental conditions and 6 non-human primates.
- Applied dynamical systems approaches to characterize the geometry of high-dimensional neural representations.
- Developed scalable preprocessing and analysis pipelines for large-scale electrophysiological datasets, improving reproducibility and robustness across recording sessions and animals.

09/2019 – 09/2020
Research Intern – Neural Signal Processing
Donders Centre for Neuroscience, Nijmegen, The Netherlands

- Applied signal processing methods and time-series modeling in electrophysiological recordings from multiple mice, evaluating cross-session and cross-subject neural variability.

03/2018 – 06/2018 **Data Science Intern – Medical Physics**
Istituto Romagnolo per lo Studio dei Tumori, Meldola, Italy

- Developed a GUI to facilitate quality assurance of particle accelerators used in radiation therapy.
- Designed protocols to predict accelerator malfunction.

Publications

2026 Reciprocal Fronto-Parietal Interactions Support Motor Anticipation during Sequential Reaching.
G. Bardanikas, ..., A. Brovelli, T. Brochier, under review in **Nature Communications**, bioRxiv: https://doi.org/10.64898/2026.03.03.709252

Premotor Cortical Interactions During Action Selection under Social Uncertainty.
G. Bardanikas, ..., I. Kagan, in preparation

Dynamic Encoding of Information Gain in Frontal Areas across Reward-Guided Learning.
G. Bardanikas, ..., E. Procyk, A. Brovelli, in preparation

Education

2021 – 2025 **PhD in Computational Neuroscience**, Aix-Marseille University, Marseille, France

2018 – 2020 **MSc in Neuroscience**, Radboud University, Nijmegen, The Netherlands

2013 – 2018 **Ptychio Physikis (Degree in Physics)**, Aristotle University of Thessaloniki, Greece

Languages

Greek Native User
English Proficient User (C2)
French Independent User (B2)
Spanish Basic User (A2)